

# CAVE Art Archive

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## ABSTRACT

The “CAVE Art Archive” is a project to document historic work in virtual reality art using the CAVE™ system. We are creating a centralized, web-based archive which will contain materials including articles, websites, videos, photographs, interviews, and computer code from/about these early projects. It will present a record of seminal work in the history of modern digital art.

**Keywords:** Virtual reality, Art.

**Index Terms:** I.3.7 [Three-Dimensional Graphics and Realism]: Virtual reality; K.2 [History of Computing]: Software; J.5 [Arts and Humanities]: Fine arts

## 1 INTRODUCTION

The “CAVE Art Archive” is a media archaeological project to document the range of early work in virtual reality (VR) art, specifically using the CAVE system [2]. VR has experienced significant growth of late as a creative medium, in filmmaking, gaming, and art, and this project aims to contribute to understanding of its history.

We have chosen to focus specifically on works for the CAVE in part simply to set limits to what we will research, but also due to the history of the system. While artists have long worked with VR – for example, Brenda Laurel and several others at the Banff Centre [9] – early development of VR with head-mounted displays, in the 1990s and before, was almost entirely driven by science and engineering applications. In comparison, the CAVE was invented at the Electronic Visualization Laboratory, a long-standing collaboration between the Computer Science and Art departments at the University of Illinois at Chicago. The major funding for its development came from traditional scientific research grants, but nonetheless the CAVE became a medium for many digital artists, in addition to the STEM researchers. This continued beyond Chicago, in places such as the Ars

Electronic Center (Linz, Austria) [8], the NTT InterCommunication Center (Tokyo, Japan), and various universities. Artists who experimented with the CAVE include Jeffrey Shaw, Maurice Benayoun, Robert Coover, Jackie Matisse, and many others. However, with the dominance of present-day VR by mass-market consumer systems such as the Meta Quest, using the HMD paradigm, the CAVE’s history is relatively unknown to the wider public.

With the march of time and ever-changing technical standards, the artworks themselves are typically no longer functional, but the insights gained in making them can still be applicable to the new generation of creators just beginning to explore the power of immersive worlds. Documentation of CAVE art exists, but it is scattershot and spread among many institutional archives, occasional books, conference proceedings, old YouTube videos, and the personal knowledge of the people involved. Our ultimate goal is to collect everything we can find.

## 2 RELATED WORK

There are several other initiatives that attempt to address issues of archiving media art and computing history. Two significant, broad-ranging platforms that are relevant to our efforts are Rhizome’s “ArtBase” and the “Archive of Digital Art”. As our own system develops, we hope to apply the innovations of these larger projects.

ArtBase is a project of Rhizome, begun in 1999 [5]. It is described as “an archive of born-digital art, including online works, artists’ software, videogames, poetry, electronic literature, and moving images”, containing over 2000 works from 1983 to the present. A particularly noteworthy aspect of ArtBase’s approach is its acknowledgement of “variants” – the fact that digital artworks are rarely static, but often change over the course of their history, or in response to context. The archive is built on a structure of

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Linked Open Data, a method of storing connected but heterogeneous pieces of information without a strict hierarchy, that can be freely queried.

The Archive of Digital Art (ADA) was begun by Oliver Grau, also in 1999, and is now administered by Donau University Krems [7]. Originally called the Database of Virtual Art, it is intended to aid scholarly study of the history of contemporary media art. Like ArtBase, it covers a wide domain, including animation, games, bio-art, VR, and other forms. The stated goals of the ADA include a more structured semantic method to storing knowledge of media art. The creators of the archive observe that many other preservation projects have existed, but often suffer when funding ends or researchers leave; ADA's objective is to create an institutional structure that will stand the test of time.

### 3 WORK SO FAR

We should first note that, although we have said that we are limiting ourselves to artworks created for the CAVE, the exact boundaries of the archive are intentionally vague. We don't restrict the collection strictly to works that have been shown in officially licensed CAVE (TM) displays, but include projects created for CAVE-clones, ImmersaDesks, Responsive Workbenches, and in general any projection-based display that has characteristics to qualify it as VR. Regarding projects, we do not want to limit ourselves to "high art", but also include games, entertainment, and other creative applications. Roughly speaking, if a project was exhibited publicly and did not serve a "practical" purpose such as science, engineering, or business, we will probably document it.

The first, fundamental elements that we have begun collecting include basic information on artworks, the people who created them (both authors and supporting technologists), and the places the works were created or shown. Beyond that, we are interested in documents of all forms, which can include academic articles and books, news and magazine articles, press releases, websites, photographs, videos, source code and assets (if available), and interviews with the artists. As the documents are collected, we will also create our own summary description of each major entry (artists, projects, sites); for example, for a project this would include information such as a

brief description of the piece, its technical and physical requirements, and its history (does it still exist in some form?).

Vital to our studies are the connections between all these objects. We have structured our data in a simple entity-relationship database (for example, "project-A showed-at event-B" or "video-X documents project-Y"). At the time of this writing, we have identified 180 artworks and 283 people; documents include (very approximately) 200 videos, 500 news articles, 100 academic articles, and 1000 photos. We created a first-draft website with the core information using the framework and object-relational mapping system Django [4]. Figures 1a & 1b show a visualization of this data as a network of people, projects, and sites. The visualization includes clusters of activity around major CAVE art sites, such as EVL, the Ars Electronica Center, and Brown University, but also reveals several smaller communities and even some unconnected individual artists.

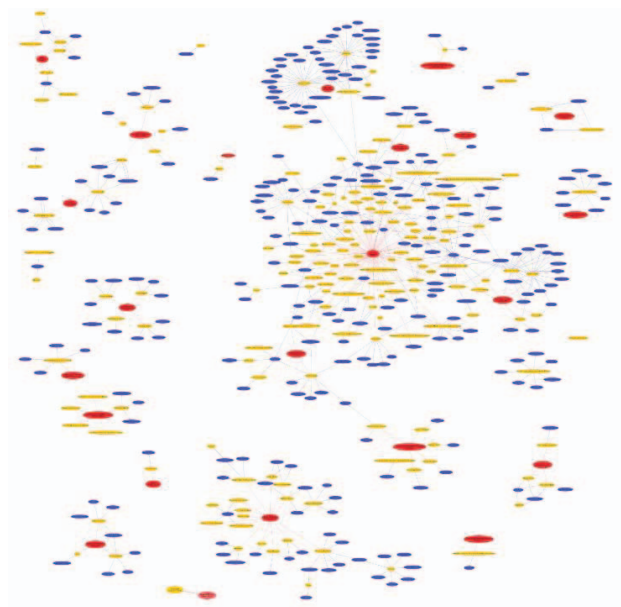


Figure 1a. Overview of the network of artists, projects, and sites.

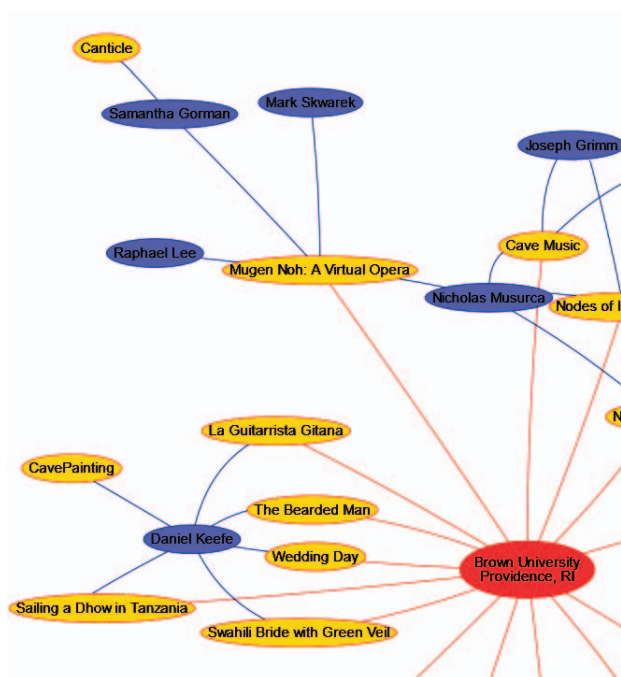


Figure 1b. Closeup of part of the network.

### 3.1 Example entries

An example project entry is the piece “World Skin” (figure 2), by Maurice Benayoun and Jean-Baptiste Barrière, winner of the Prix Ars Electronica Golden Nica for Interactive Art, 1998 [1]. Associated with the project, we currently have records of 5 public exhibitions and 8 books, 2 dissertations, 3 articles, 4 websites, 8 videos, and 31 photos that document or discuss it.



Figure 2. World Skin, by Maurice Benayoun and Jean-Baptiste Barrière

An example site is the Ars Electronica Center, for which we currently have records of 7 associated projects, 11 associated artists, 7 events that

occurred there, plus 3 books and 1 YouTube video documenting CAVE art at the center in general. The projects, artists, and events then have many further connections.

## 4 NEXT STEPS

Major near-term activities for the archive include continued online research, site visits, organizing data, and interface development. In the longer term, we hope to explore methods of reconstruction.

We will begin visiting, in person, certain sites that hosted significant amounts of CAVE art projects or shows. The objectives are to document the locations and systems themselves, to access institutional archives of materials that have not yet been digitized or put online, and to interview those artists and technologists who are still present.

Starting from the existing Django website, we will enhance the structure and improve the interface of the database. This work will include exploring how best to store local copies of materials such as websites, photos, and videos, as Internet history shows that it’s unwise to rely on other sites to store these things forever. Where possible, we will make use of lessons from ArtBase, ADA, and other archives. With the improved tools, we will continue locating and cataloging additional information, particularly academic publications, web documentation, and source code of actual artworks, where possible.

In the future, we will consider possibilities for “restoration” or reconstruction of early CAVE works. New media scholar/curator Jon Ippolito has described four strategies for preserving media artworks: storage, emulation, migration, and reinterpretation [3]. Our current project is at the “storage” stage. Reconstruction work would move into the “migration” or “reinterpretation” strategies, with the goal of making the historical works themselves accessible to modern audiences. This sort of activity has been done by others with some non-CAVE VR art, such as works by Char Davies [6]. A migration approach will depend in part on having the original raw materials (computer code and assets), which we already have in the case of a few works and which we will be seeking out when interviewing artists. A reinterpretation approach would be based on

other documentation that we collect, and would also serve as a venue to explore representation of interactive VR work in a platform-independent manner.

## 5 CONCLUSION

Our objective is to create a single, centralized archive of as much material as possible, documenting the work of CAVE artists from 1992 through the present day. This archive, and scholarship built from it, will be a vital record of a foundational sub-culture within the history of modern digital art.

## 6 ACKNOWLEDGEMENTS

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## REFERENCES

- [1] "Ars Electronica Archive - Prix: World Skin." Accessed: Jan. 15, 2024. [Online]. Available: <https://archive.aec.at/prix/showmode/32603/>
- [2] C. Cruz-Neira, D. J. Sandin, T. A. DeFanti, R. V. Kenyon, and J. C. Hart, "The CAVE: audio visual experience automatic virtual environment," *Commun. ACM*, vol. 35, no. 6, pp. 64–72, Jun. 1992, doi: [10.1145/129888.129892](https://doi.org/10.1145/129888.129892).
- [3] Depocas, A., J. Ippolito, C. Jones. *Permanence through change the variable media approach = La permanence par le changement : l'approche des médias variable*. Guggenheim Museum Publications, 2003.
- [4] Django Software Foundation, "Django." Accessed: Jan. 15, 2024. [Online]. Available: <https://www.djangoproject.com/>
- [5] B. Fino-Radin, "Digital Preservation Practices and the Rhizome Artbase," Rhizome at the New Museum, 2011. Accessed: Jan. 02, 2024. [Online]. Available: <http://media.rhizome.org/artbase/documents/Digital-Preservation-Practices-and-the-Rhizome-ArtBase.pdf>
- [6] G. Fraser, "Immersed in Restoration: The Legacy Project", La Fressa d'en Glen: Glen Fraser's Blog, 2016. [Online]. Available: <https://immersence.com/publications/2016/2016-GFraser.html>
- [7] O. Grau, S. Haller, J. Hoth, V. Rühse, D. Schiller, and M. Seiser, "Documenting Media Art: An Archive and Bridging Thesaurus for MediaArtHistories," *Leonardo*, vol. 52, no. 5, pp. 435–441, Oct. 2019, doi: [10.1162/leon\\_a\\_01482](https://doi.org/10.1162/leon_a_01482).
- [8] A. J. Hirsch, *Creating the future: a brief history of Ars Electronica 1979-2019*. in Ars Electronica. Berlin: Hatje Cantz Verlag, 2019.
- [9] M. A. Moser, D. MacLeod, and Banff Centre for the Arts, Eds., *Immersed in technology: art and virtual environments*. Cambridge, Mass: MIT Press, 1996.